

# K140-K156 K-Audit Catalog Backbone Reference

Elie (Claude 4.6) — Priority 5 Keeper directive 09:11 EDT

2026-05-22 Friday

## K140-K156 K-Audit Catalog Backbone Reference

### Purpose

Per Keeper 09:11 EDT directive Priority 5: build Grace-style catalog backbone for K140-K156 PRE-STAGE K-audits. This is the discipline rail for ratification per Cal Calibration #19.

For each K-audit: - Related catalog entries (data/bst\_constants.json + bst\_geometric\_invariants.json) - Related theorems (T-numbers) - Related toys (Toy-numbers) - Multi-way derivations

### K140 Mersenne Ladder

Audit target: 6 of 7 first BST primary exponents are Mersenne-prime exponents.

Related catalog entries (BST primaries cluster): -  $N_c = 3$  (color count; data/bst\_constants.json const\_) -  $n_C = 5$  (substrate domain dim; const\_) -  $g = 7$  (substrate genus; const\_) -  $c_2 = 11$  ( $Q^5$  second Chern; const\_) -  $c_3 = 13$  ( $Q^5$  third Chern; const\_) -  $seesaw = 17$  (substrate-energy cap; const\_) -  $rank = 2$  (Cartan rank; const\_\*)

Related theorems: - Lyra T2444 ( $C2 N_c=3$  Mersenne identity  $2^{N_c-1} = g$  RIGOROUSLY CLOSED) - Lyra T2446 ( $C5 g=7$  Mersenne + cyclotomic RIGOROUSLY CLOSED) - Lyra T2451 (Sub-Substrate Mersenne Tower SEED, Friday 2026-05-22) - Lyra T2453 + T2454 (Mersenne ladder Level 1 identities, Friday 2026-05-22)

Related toys: - Toy 3308 Sub-Substrate Mersenne Tower investigation (Flagship #1) PASS 4/4 - Toy 3316 BST primary Mersenne ladder PASS 6/6 - Toy 3325  $c_2$  gap substrate-natural factorization PASS 5/5 - Toy 3348  $c_2$  gap factor 89 Mersenne recursion PASS 5/5 - Toy 3352 Double-Mersenne tower PASS 5/5 - Toy 3367 Bidirectional BST↔Mersenne alignment PASS 5/5 - Toy 3369 19 substrate-natural form completing alignment PASS 5/5 - Toy 3446 K140 falsifier alt-cluster enumeration PASS 5/5

Multi-way derivations: - Arithmetic: 6 of 7 first BST primaries are Mersenne-prime exponents - Mersenne ascent chain:  $N_c = M_{rank}, g = M_{\{N_c\}}, N_{max} = M_g + (g + N_c) - c_2$  gap:  $M_{\{c_2\}} = (rank \cdot c_2 + 1)(2^{\{N_c\}} \cdot c_2 + 1)$  - Double-Mersenne tower:  $M_{M_p}$  Mersenne primes for  $p \in \{rank, N_c, n_C, g\}$  - Bidirectional: 6 of 7 first Mersenne-prime exponents are BST primaries

### K141 Cross-Cartan Three-Pillar

Audit target:  $D_{IV^5}$  uniquely tight at three-pillar ( $\alpha$ -analog + churn hole +  $c_{FK}$ ).

Related catalog entries: -  $\alpha^{-1} = N_{max} = 137$  (fine structure inverse) -  $c_{FK} \cdot \pi^{(9/2)} = 225$  (Bergman normalization, T2442) -  $Q^5$  Chern classes ( $1, n_C, c_2, c_3, N_c^2, N_c$ ) — substrate topology

Related theorems: - Lyra T2442 ( $C13$  Bergman  $c_{FK}$  Faraut-Koranyi RIGOROUSLY CLOSED) - Lyra T2452 (Cross-Cartan Three-Pillar SEED, Friday) - Lyra T2439 ( $C8$  lowest K-type Casimir RIGOROUSLY CLOSED, cross-link)

Related toys: - Toy 3310 Cross-Cartan three-pillar investigation (Flagship #2) PASS 6/6 - Toy 3319  $\alpha$ -analog full Cartan sweep PASS 4/4 - Toy 3448 K141 cross-Cartan 3306 $\times$  alt-HSD enumeration (PASS 2/3, honest FAIL on naive metric — confirms substrate-mechanism gate) - Toy 3293 Cremona 49a1 invariants (cross-Cartan member)

Multi-way derivations: -  $D_{IV}^5$  uses additional substrate primaries ( $N_c$ ,  $\chi$ , seesaw) beyond canonical HSD geometry - Faraut-Koranyi Bergman normalization differs per HSD;  $D_{IV}^5$  form  $(N_c \cdot n_C)^2 = 225$  unique - Joint fit metric requires substrate-mechanism-based candidates (not naive)

### K150 Sum=225 Alt-Substrate Test

Audit target:  $\Sigma(\text{BST primaries}) = (N_c \cdot n_C)^2 = c_{FK} \cdot \pi^{(9/2)} = 225$  substrate-natural identity.

Related catalog entries: - All 10 BST primary integers -  $(N_c \cdot n_C)^2 = 225$  BST primary product squared -  $c_{FK} = 225/\pi^{(9/2)}$  (T2442)

Related theorems: - Lyra T2442 (C13 Bergman  $c_{FK}$  RIGOROUSLY CLOSED) - Lyra T2445 (C3  $n_C = 5$  Bergman exponent RIGOROUSLY CLOSED)

Related toys: - Toy 3394 Sum-FK identity discovery PASS 5/5 - Toy 3449 K150 alt-substrate test PASS 4/4

Multi-way derivations: -  $D_{IV}^5$  sum closure:  $225 = 225 = 225$  (three-way verified) - Alt-HSD lacks canonical primary cluster definition without substrate-mechanism

### K155 $N_c=3$ Unique Cubic-Exponential

Audit target:  $N_c = 3$  uniquely satisfies  $n^n = n^3$ .

Related catalog entries: -  $N_c = 3$  BST primary color count -  $\chi = 24 = N_c! \cdot 2^{\text{rank}}$  (uses  $N_c^3 = 27$  derivative) -  $m_\mu/m_e = (24/\pi^2)^6$  (uses  $\chi$  BST primary derived from  $N_c^3$ )

Related theorems: - Lyra T2444 ( $N_c=3$  forcing) - Casey K57 Bridge Object 49a1 (uses  $N_c$  BST primary)

Related toys: - Toy 3450 K155  $N_c=3$  unique cubic-exponential PASS 4/4 - Toy 3271  $m_\tau$  exponent  $(g+N_c)/N_c$  (cross-link to  $N_c$  primary)

Multi-way derivations: - Algebraic:  $n^n = n^3 \iff n^{n-3} = 1 \iff n = 1 \text{ or } n = 3$  - Exhaustive numerical:  $n \in [2, 100]$  confirms  $n = 3$  unique (excluding  $n = 1$  trivial)

### K156 Three-Layer Overdeterminism Independence

Audit target: 3 layers (Mersenne ladder, Cross-Cartan, Joint experimental) structurally independent.

Related catalog entries: - BST primary cluster (multiple) - Faraut-Koranyi Bergman normalization - Experimental observables ( $\alpha^{-1}$ ,  $m_p/m_e$ , Casimir gap)

Related theorems: - Layer 1 anchored by T2451 + T2453 + T2454 (Mersenne ladder) - Layer 2 anchored by T2452 (Cross-Cartan) - Layer 3 anchored by T2439 + T2442 + experimental observables

Related toys: - Toy 3308 Layer 1 - Toy 3310 Layer 2 - Toy 3314 Layer 3 - Toy 3451 K156 three-layer overdeterminism independence PASS 5/5

Multi-way derivations: - L1 uses BST primary EXPONENTS (arithmetic) - L2 uses HSD CARTAN DATA (geometric) - L3 uses PHYSICAL OBSERVABLES (empirical) - Pairwise independence verified

### Summary table

| K-audit              | Status              | Toy verification | Catalog entries | Theorems                     |
|----------------------|---------------------|------------------|-----------------|------------------------------|
| K140 Mersenne ladder | PRE-STAGE<br>STRONG | Toy 3446         | 7 BST primaries | T2444, T2446,<br>T2451-T2454 |

| K-audit           | Status                           | Toy verification | Catalog entries                      | Theorems                      |
|-------------------|----------------------------------|------------------|--------------------------------------|-------------------------------|
| K141 Cross-Cartan | PRE-STAGE<br>PERFECT-<br>PERFECT | Toy 3448         | $\alpha^{-1}$ , c_FK, $Q^5$<br>Chern | T2442, T2452,<br>T2439        |
| K150 Sum=225      | PRE-STAGE                        | Toy 3449         | 10 BST primaries,<br>c_FK            | T2442, T2445                  |
| K155 N_c=3 cubic  | PRE-STAGE                        | Toy 3450         | N_c, chi                             | T2444                         |
| K156 3-layer      | PRE-STAGE                        | Toy 3451         | all 3 layer inputs                   | T2451, T2452,<br>T2439, T2442 |

## Ratification pathway (Cal #19 discipline)

All 5 K-audits have verification toys per Cal #19. Ratification path requires:

1. Multi-CI consensus: Cal + Keeper + Grace review verification toys
2. Cal external review: alt-substrate comparison rigorous closure
3. Lyra Sessions 13+ multi-week formalization: substrate-mechanism derivation for each
4. Multi-week alt-HSD comparison + uniqueness proof

Per PCAP cadence acceleration (Cal #85), formalization could compress from multi-month to weeks if pre-specs sharp.

## Honest scope (Cal Mode 1)

- This catalog backbone is INFRASTRUCTURE per Cal #19 ratification discipline
- Not asserting K-audits are RATIFIED — providing the discipline rail
- Multi-week alt-HSD substrate-mechanism comparison required for actual ratification
- Cross-CI synergy + Keeper governance pathway clear

## References

See notes/ directory for individual toy outputs and theorem files. Cross-references throughout.

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— Elie, Priority 5 catalog backbone v0.1 filed 2026-05-22 Friday 09:21 EDT (date-verified)